



AI in Sports: A Machine Learning Approach for Improving Video Analysis in Soccer

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Abstract—As artificial intelligence (AI) continues to grow in computational prowess, it has begun to see mainstream usage for both personal and industry use. One such industry that has benefited from this surge is data analysis. Data, in general, is the core of Artificial Intelligence, especially in its subfield of Machine Learning, in which it attempts to identify patterns on given information. Noting this, there is a major industry that could leverage a combination of machine learning (ML) and data analysis: sports. In order to continue improving their own performances, many teams and/or clubs typically analyze their previous outings as a barometer of how well they are playing. In the world of soccer, many professional outfits go as far as to hire data analysts to peruse over this data and relay information to the head coaches and tacticians. However, carrying out the analysis is much easier said than done; it can be a difficult and daunting task, especially if the medium is video. In this project, we explored the avenues of how ML models could aid in efficient analysis via real time player and action detection to provide comprehensive statistics for performance assessment.

I. INTRODUCTION

At the moment, it would not be incorrect to say we have arrived in the great age of AI. Technological advancements are virtually made everyday. Modern AI models have evolved from performing mathematics [1] and menial tasks to generating art [2] and creating music. [3] AI is now continuously being used to solve increasingly complex problems on every corner. A good portion of this progress is made possible through the power of ML. The goal of machine learning is to learn patterns present within data that it is shown. This task is typically performed using neural networks, an architecture

modelled after neurons in the brain, that feed information through its various layers in efforts to give the output that it was assigned to provide.

Neural networks are behind a good number of recent AI developments. It has aided in the creation of text generation, image classification, and object recognition. Many of these tasks that ML networks perform lean into specialized industries. One such industry is sports, where AI is used to analyze performances and how well a team may perform in the future. In general, this form of data analysis is generally useful for professional teams. Going over their past victories and defeats gives the team a good insight into where and how they can improve tactically. In the sport of soccer, for example, the main medium they use to peruse through this data is video. Statistics alone don't always tell the full picture; combined with recordings of their previous outings, they can better understand where the numbers come from. Due to the importance of this internal research, many soccer clubs employ individuals and/or entire teams to look through their copious amounts of visual information. Despite its usefulness, it is not an easy process. Soccer is not a short game, by regulation matches are 90 minutes long with upwards of 20 minutes of stoppage time played in each half. In addition, data analysts would need to track 22 players on the pitch, follow the ball throughout play, and ensure they correctly identify events.

We believe that automation could help improve this intensive process, whilst keeping it as accurate as possible. In this project, we explored the different ways that AI can be used to analyze footage of soccer games and provide useful insight into the statistics present within the frames of video.

Note that the header image was *generated* by AI from the prompt "An abstract painting of Robots playing soccer".

Utilizing object tracking, pose estimation, and much more, we discovered how to detect events such as goals and set pieces, track who is in control of the ball, and understand the motion of players on the field.

II. BACKGROUND

A. Previous Work

Efforts to combine AI with sports have not been uncommon. It has been applied to assess performances for weight training [4] and even the creation of soccer-playing robots for competitive use. Further in the world of data analysis, researchers have explored the possibilities of using AI for analyzing the playing styles of different soccer teams [5] and ML models have been used for hybrid architectures that provide visualization for sports data. [6] In terms of video analysis, there has been previous research in which AI techniques are used to create highlight reels of soccer matches based on detected events, logos, and camera angle changes. [7]

B. Techniques in AI

We chose to follow modern techniques and use a neural network as the crutch of this project. As stated previously, neural networks are an ML architecture that attempts to learn patterns from previously seen data. It learns this information by adjusting weights attached to each of its "neurons". This essentially creates a mathematical equation that best fits the defined problem and provides correct output as consistently as possible. This machine learning application has many derivatives, including convolutional neural networks (CNN). [8] CNNs are often used when dealing with image and video data due to its feature extracting capabilities. One such CNN that is commonly used is You Only Look Once (YOLO). YOLO in particular allows for real time detection of objects of interest, which led us to use it as the core of our project. [9]

Another commonly used technique when dealing with image and video data is edge detection. [10] This form of detection searches for lines throughout an image based on a color threshold. Once it finishes gathering the information from the image, it translates the lines into a binary image with the detected contours being the activated pixels. There are many programming libraries that can perform this form of image analysis, one being OpenCV. [11] This library of computer vision tools provides image manipulation methods for scientists to understand the complexities lying within a frame of video.

In the world of object detection, there are some additional techniques that for allow us to improve the tracking of objects. At certain points of a video, a ML model may not be able to correctly identify a specific object, making it difficult to track its position, especially if you are mathematically storing it. One such technique is a Kalman filter. [12] The filter will predict the next location based upon the calculated movement of the object from the previous positions as input.

It is also important to note that there are more advanced techniques within the world of predicting objects, especially

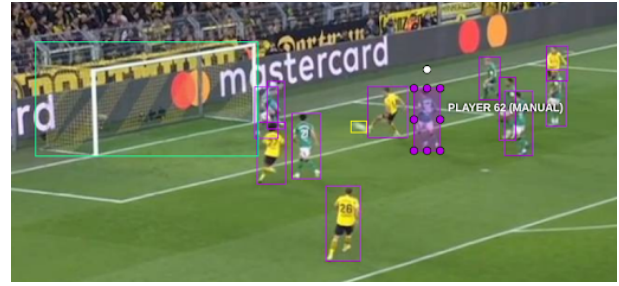


Fig. 1. Example of annotating the frames of video for the YOLO model to train on.

ones as complex as humans. For example, if one wanted to identify the current stance a person is taking within an image or video, they may want to use pose estimation. [13] Pose estimation involves following the movement of points of a person and/or object. These models aim to understand the spatial positioning of these key points and provide researchers with coordinates, three or two dimensional, for further use.

III. DATA

Due to being unable to physically go to football matches, we opted to use broadcast game footage instead. Over the following weeks, 95 videos were collected from the following competitions:

- 1) Premier League (England)
- 2) La Liga (Spain)
- 3) Bundesliga (Germany)
- 4) Eredivise (Netherlands)
- 5) UEFA Champions League (All European Countries)
- 6) Copa Libertadores (All South American Countries)
- 7) Saudi Pro League (Saudi Arabia)

The data was collected with the notion of diversity. We ensured that the video footage contained a vast number of football events to pick apart and use as training data for our CNNs. The events in question included goals, set pieces, shots, and passes. Matches that had a good probability of seeing a large numbers of these, often high profile affairs, were carefully chosen as we could not record more than two games at once due to hardware limitations.

As for storing the various video data, we placed them into separate hard drives due to the large file size that the footage ended up amounting to. They were named according to the date it was recorded and the participating teams.

IV. METHODS

A. Detecting Objects on the Field

In order to train the model to detect objects, we had to tell the model what part of each frame of video we are interested in. To perform this, we used Computer Vision Annotation Tool (CVAT) [14], a web application that allowed us to mark specific areas of an image that we wanted the machine learning model to train on.

These key parts in particular are:

- Players

They are our actors and their movements define most statistical measurements for soccer.

- Ball
The game is centered around this object.
- Net
For detecting goals and which player is the goalkeeper (for set pieces).
- Corner Flag
For the contextual event of a corner kick.

As you can see in Figure 1, the areas of the image containing these objects and persons are annotated using boxes. Once the annotations of the video frames are completed, they are exported into a text format that the YOLO model interprets as features of interest.

B. Calculating Ball Possession

Now with the ability to track the players, we used this information to track the amount of ball possession a team has within footage. Ball possession is useful for teams to know how well their players can retain the ball and recover it from opponents. [15] In order to calculate who has the ball at each frame of video, we identify which team the player closest to the ball plays for.

In order to do this, we naively assume the first player closest to the ball in the video is from the team specified by the user. After getting this information from the first frame, we begin our loop.

- 1) Get the bounding box of the player closest to the ball.
This is done by getting the center coordinates of the relevant bounding boxes and calculating the Euclidean distance between them.
- 2) Crop the player out of the image and get their hue histogram.
This is done programmatically using computer vision libraries.
- 3) Compare the histogram to the histogram of the previously closest player.
If the distributions of color are at least 50% similar, we consider that a player on the same team.
- 4) Switch the team currently in possession accordingly.
This is as simple as changing the current team ID within the program.

C. Detecting the Area of Play

Expanding upon understanding the movement of the ball between players, we also wanted to understand how we could track the location of play on the pitch. Combined with possession, knowing which part of the pitch a player has possession of the ball can provide useful information about a teams attacking threat and defensive capabilities.

For example, if a team has possession of the ball in their opponent's half for an extended period of the game, that may tell us that they are dominating the match through their attack. However, if you further acknowledge statistics such as a low shot count, that may mean that the opponent's defense is very well organized.

Although some coaches may split the field into upwards of 20 zones, in order to simplify the problem into an expandable state, we only concerned ourselves with the main 3 field portions that encompass these 20 areas.

- 1) Home Attacking Third
- 2) Middle of the Park
- 3) Away Attacking Third

As shown in Figure 2, the two attacking thirds contain the opposing teams' penalty boxes (and the surrounding area of said boxes) whilst the middle of the park pertains to the halfway line and the center circle in which the game is kicked off from.

We leveraged to use another custom CNN in order to ensure we extracted all of the context needed from each training image as well. This neural network was much less complex than that of YOLO and other previously seen due to the images being binary.

The training images in question were not just the raw images of the pitch sections, however. We were not concerned with the colors of the players, the field, or anything of that sort. We wanted the model to understand the lines that make up the thirds of the playing field. To ensure this, we used edge detection to extract the lines detected lines on the pitch and place it into a binary image that highlighted them. From there, each image was given its according label and passed into the model for training and testing.

D. Detecting a Goal

In order to detect a goal, we first needed to break down how a human would decide whether or not a goal is awarded to a specific team. A goal is of course given once the ball crosses the goal line and enters the net. We decided to simplify this down to when the ball "collides" with the net.

E. Detecting Set Pieces

Set pieces are points of play that are awarded when the ball is out of play or a foul is called. There are various types but the ones we are most concerned with are:

- Free Kicks
Awarded outside of a penalty box. The opposing team is permitted to setup a wall of players to directly block the path of the ball. It is important to note that depending on distance a wall is not always present.
- Penalties
Awarded inside of a penalty box if a foul occurs within its boundaries. The match ball is placed on the penalty spot and player takes a shot against the sole goalkeeper.
- Corner Kicks
Awarded when the ball crosses the line parallel to the penalty boxes net. This preliminary action must have been performed by team who "owns" the net.
- Goal Kicks
Awarded when the ball crosses the line parallel to the penalty boxes net. This preliminary action must have been performed by team who does not "own" the net.

Examples of these four set pieces are shown in Figure 3.



Fig. 2. Examples of the different areas of play that a broadcast can be focused on, such as the home attacking third (left), middle of the park (middle), and the away attacking third (right).



Fig. 3. Examples of set pieces that can occur during a soccer match. Shown are a direct free kick (top left), a penalty kick (top right), a goal kick (bottom left), and a corner kick (bottom right).

In order to know when during a video that a set piece is occurring, we must determine if there is a "dead ball" sequence occurring at a various times.

What we define as a "dead ball" situation is when the elements on the pitch have not moved a large distance in some time. From here, we assume that the ball is not in play and a set piece is being prepared for taking.

On average, we discovered that this precedent typically occurs for 10 frames before the set piece is taken. Therefore, whilst analyzing the video, if the distance moved by the key parts has not changed dramatically for at least 10 frames, the program will flag it for review. Blessed with the information of how these set pieces were similar, we began exploration of how we could differentiate them.

1) *Free Kicks and Penalties:* Due to the similar circumstances in which these two set pieces can occur during a match, we decided to explore how we could separate them from each other.

Initially, our thoughts were to simply check if the set piece taker and the ball were within the penalty box present in the image. We decided against this due to the fact that the bounding box a YOLO model may draw around the penalty box would not be an accurate depiction of the area. Due to the angle that the broadcast cameras capture the field, small parts of the surrounding area may end up being included in the collision detection and lead to blatantly incorrect predictions of a penalty kick.

Instead, we realized that we could separate a free kick from

a penalty by using collisions of our original key elements, rather than the place of play. One of the main discerning features of these two set pieces is the presence of a group of players blocking the path of the ball. Due to soccer regulation, players are not allowed to obstruct a penalty kick, they must stand behind the taker at all times. Meanwhile for free kicks, even in cases where a team has chosen to not form a typical wall, there is generally several players that intersect the path to goal.

With this information, we can detect whether or not a set piece is a free kick by simply "drawing" a line from the center coordinates of the ball to the net. If there are any collisions (excluding the goalkeeper and the taker), we can safely assume that a free kick is taking place. If not, we flag it as a penalty.

2) *Goal Kicks:* Similarly, we can use the idea of proximity to detect when a goal kick is occurring. As stated before, for free kicks and penalty kicks, the closest player to the ball is predicted to be the taker. However, if the taker is also closest to net (the predicted goalkeeper), we will mark this event as a goal kick.

3) *Corner Kicks:* Corner kicks, however, do not need this extra layer collision. A key defining feature of what determines this set piece is the location of the taker and ball relative to the presence of the corner flag.

Simply put, if a player and the ball are within a small pixel proximity of the corner flag during a "dead ball" scenario, we flag that as a corner kick.

F. Action Detection

It became clear from the start that a typical "box image" approach for shot detection was not going to work. Upon testing, attempting to use a frame of when the player made contact with the ball (using their foot) performed poorly, even going as far as to incorrectly identifying a group of players on the pitch as taking a shot. This had caused us to realize that these actions are best identified as a sequence of positioning rather than just one frame of positioning.

The processing of data for detecting a shot differed from the cropping of forms of interest as the previous explorations. In order to correctly identify the poses of the player(s) on the pitch, we opted to use a key-point detection system. Similar to previous research on sign language detection [16], we used identified body parts that were key to defining the overall structure of each frame of movement, also known as pose



Fig. 4. Shown is an example of how the virtual skeleton is applied to a player on the pitch. All of the relevant joints are marked, such as the shoulders (1, 2), elbows (3, 5), wrists (4, 9), hips (7, 8), knees (9, 11), ankles (10, 12), and playing ball (13).

estimation. Using CVAT's skeleton definer, we tracked key-points of players throughout several frames of video.

The key-points that we decided were the most important in deciding the player pose were the:

- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles
- Playing Ball

As you can see in Figure 4, when these joints form edges with one another, they form a human like skeleton that helps provide a near-accurate estimate of each players pose as they move across the pitch. We also included the match ball because we are most concerned with the movement and positioning of when a player is in possession. A shot or pass is only registered when a touch is made. All of these annotations were then translated into another text format for use in a custom-trained YOLO model that specifically deals with estimating the poses within an image.

V. RESULTS

A. Detecting Objects and Players

Overall, the model performed fairly well when detecting the most important parts of an image. In Figure 5, you can see an good example of a well predicted frame. Meanwhile in Figure 6, you can see an example of an occasional fault during analysis. In terms of the data it was trained on, there

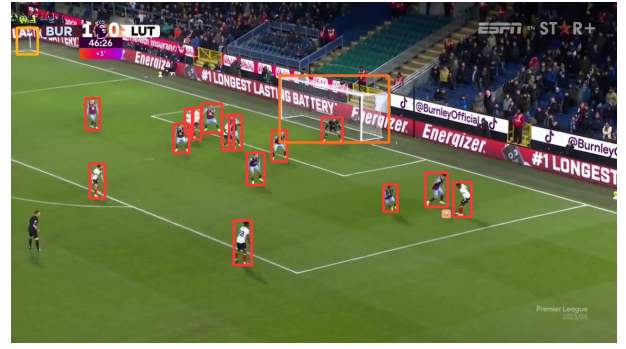


Fig. 5. Example of a good detection that can occur during the analysis of a game. All important parts of the current scenario are correctly detected.

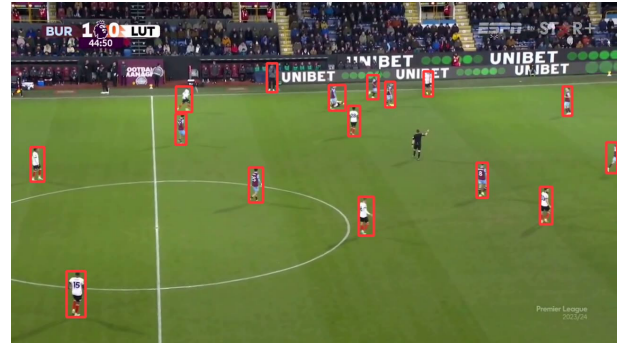


Fig. 6. Above is a depiction of some errors during analysis of a video. One of the coaches standing upon the touchline is inferred to be a player.

are two metrics that helped us understand how well the model was doing during training: precision and recall. Precision is defined by the number of correct predictions out of the number of predictions made for a specific label. Recall is defined as the number of correct inferences of a label compared to all the predictions of a label, correct or not. As the equation below show: TP stands for true positives, the correct predictions of a specific label, FP stands for false positives, the incorrect predictions that used the same specific label, and FN stands for false negatives, is the missed predictions of that label.

$$precision = \frac{TP}{TP+FP} \quad recall = \frac{TP}{TP+FN}$$

Our custom YOLO model scored averaged 97.9% precision and 97.2% recall for all possible labels.

B. Ball Possession

Initially, the detections for ball possession were not very good. We ignored any frames that did not have the ball and that led to many lopsided and flat-out incorrect calculations of the amount of possession a team had. As one could infer by this discovery, the calculation of possession is heavily dependent on consistent detections of the playing ball, but as previously stated, the model had several issues in detecting the ball in motion.

In order to mitigate this error, we decided to use the aforementioned Kalman filter. As the model parsed through the video, we would feed the seen positions in the Kalman filter. Once a frame without a detection of the ball was given,

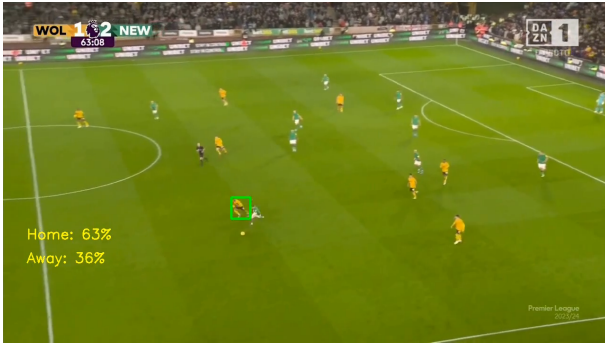


Fig. 7. Example of calculating possession using the positioning given by the YOLO model. The player considered to be closest to the ball is highlighted with a green box.

we would get the predicted position from the Kalman filter and used that to mathematically find the player closest to the ball.

By using the Kalman filter, the overall performance for detecting possession improved almost dramatically. The quality of calculations averaged around a 5% to 10% margin of error. In Figure 7, you can note that the calculated possession for that video at that current time is 64% for the home team and 36% for the away team (These numbers are rounded up for readability). The actual numbers should be 71% to 29% for the home and away sides, respectively. Although it may not be immediately apparent, this 8% margin of error is not necessarily a bad thing. Possession in soccer is not always accurate in practice. Many competitions calculate it differently, whether it be the number of passes between players, time on the ball, or number of touches. As long as it gives us a good idea of who has been in control of the ball for a portion of the game, as shown previously, it is more than adequate.

C. Location of Play

As stated in Section IV, we performed edge detection on images of the locations of play in order for training on the models. However, as shown in Figure 8, the original colored images must first be translated into grayscale before passing it into OpenCV's edge detection algorithm. This allows the intensities of the lines to be more apparent to the edge detector.

Summarily, the CNN trained very well on the edge data. On the training set it was given, the algorithm successfully learned all of its images. The testing set, however, only scored around 70%. This may seem underwhelming at first, but looking under the surface it was easy to tell where the neural network was making mistakes. The 3 portions of the pitch, especially the 2 attacking thirds, contain very similar shapes that gave the model difficulties. In fact, the most areas of the pitch that the model had the hardest time differentiating from was the two attacking thirds. We suspect that this is mainly due to the only discerning feature between these two parts of the pitch are the fact that they are naturally flipped.

D. Goals

As shown in Figure 9, our architecture is proven to be able to detect goals. However, this does contain some caveats. Due to the way the collision between the net and goal are detected, there are many false positives. This occurs because the model has no concept of depth; it simply checks if the two-dimensional bounding boxes of both detections intersect at any point.

E. Set Pieces

As shown in Figures 10, 11, 12, and 13, the scenario detection for set pieces was quite successful. The frame average of 10 allowed for almost every instance of set piece to be detected. The program only struggled to properly detect a set piece in general when the cutting of the footage caused transitions from dead balls to set pieces to be very short. To elaborate, many broadcasters attempt to "fill in" the time in-between dead balls and active play by having a plethora of player close-ups and replays of previous events. However, the buffer to check for dead-balls can be easily adjusted by the user and applied on a case-by-case basis.

Furthermore, the only category of set piece that provided any true struggles in detection were corner kicks. Due to the small number of pixels that represent corner flags in each frame of video, the model had a hard time detecting the corner flag at the same time it would detect the ball and the player closest to the corner flag.

F. Action Detection

As previously mentioned, we used the concept of pose estimation to properly understand how a player moves within a specific action. After applying the custom skeleton to different players across thousands of images, we ended up encouraging results depicted in Figure 14. These model that provided those results scored 99.9% precision and 100% recall.

Unfortunately, pose estimation came with a myriad of difficulties. Due to the low resolution of the video, the YOLO pose model had issues attempting to consistently detect the player with the ball. The low amount of pixels would often given the model problems when trying to differentiate between the forms of the body. In addition, due to the speed in which players move, the footage would smear and cause further manipulation that made the players unrecognizable. The movement of the broadcasting camera also proved to be a catalyst for this as the constant zooming caused the sizing of pitch elements to be inconsistent.

As such, many frames had empty predictions and using a Kalman filter was proven to be futile. Although it would sometimes correctly label the player with the ball, at times the skeleton orientation was blatantly incorrect. It would only cause more problems attempting to predict key point locations based on false data since it would cause the pose estimations to spiral into instability. To properly understand the poses for sequences and possibly use another model to predict sequence type based upon these estimations, we need them to be as accurate as possible.



Fig. 8. Showcase of the steps required to perform edge detection. The raw image (left) is translated into grayscale (middle) and OpenCV's Canny edge detection is applied to find the pitch markings (right).

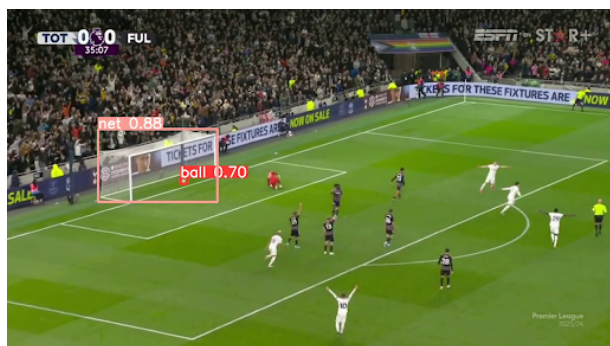


Fig. 9. Example of a correct goal prediction. The bounding box of the ball is within that of the net.



Fig. 12. Example of a correct goal kick prediction. Notice how the goalkeeper is the closest person the ball.

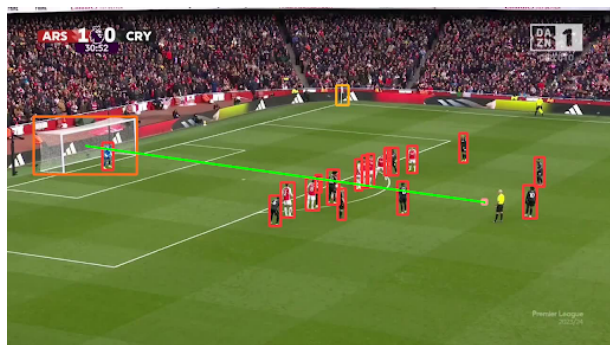


Fig. 10. Example of a correct free kick prediction. Notice how bounding boxes of the players are colliding with the standing on the penalty box line collide with the inferred direction of the free kick.

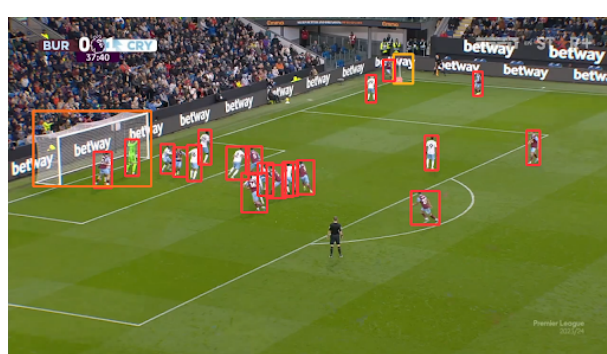


Fig. 13. Example of a correct corner kick prediction. The taker and the ball are within close pixel proximity of the corner flag.

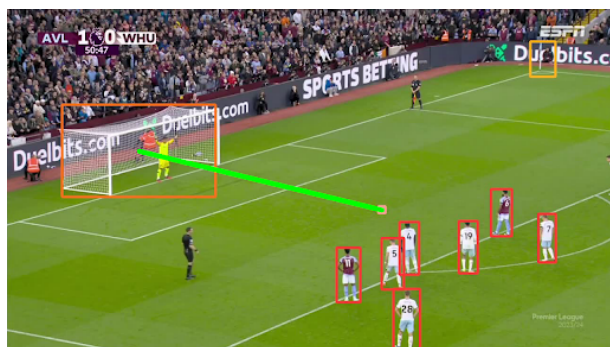


Fig. 11. Example of a correct penalty kick prediction. Notice how there are no players within the penalty box that can obstruct the line from the ball to the net.

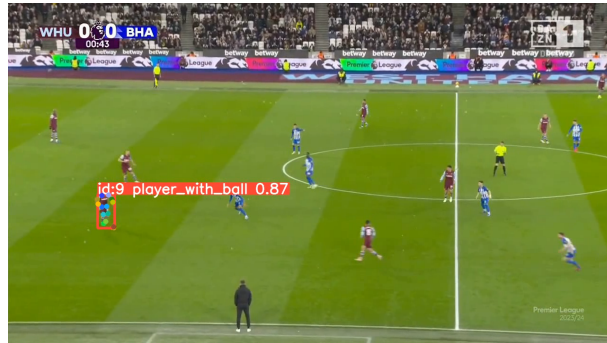


Fig. 14. Example of action detection on a live video. The colored dots that cover the joints of player denote the key points used to detect his current pose.

VI. FUTURE WORK

After conducting these various explorations, there are several points of interest in this research where further developments can be made.

A. Projecting 2D to 3D

Based on our research in detecting goals and possession, we believe that 3D projection could be utilized to further understand the depth within the image. This does not necessarily require extra software to be running alongside the model, this could be performed by employing non-trivial mathematics. For example, if we projected the posts and outer lines of the net onto a 3D plane, it would give us a better understanding of whether a ball entered the net or not. Furthermore, this would allow us to better calculate the distances between different elements on the pitch. Currently, we calculate distances by getting the euclidean pixel distance between the centers of each object's bounding box. This can sometimes be faulty because of the angle matches are typically broadcast at; some players may seem closer than what they really are. With 3D coordinates, we would get a much better spatial understanding of where the bounding boxes stand and allow us to more accurately get statistics such as possession and, to an even further extent, set pieces. Currently, only one line is drawn in order to denote the possible path the ball can make. This is mainly because too many lines may include players that are not in the path of the ball at all. If the field was brought into 3D, we could form a cone of lines that provide a good range of where the ball could land. With this, we could have a better spatial understanding of whether or not a player collision is occurring within the set piece area.

B. In-Person Footage Collection

Due to the issues that we experienced with low resolution footage, a glaring improvement would be to collect higher quality footage. In the future, we would prefer to personally attend matches and acquire footage on the sidelines. The location of the sidelines is important because it would allow us to much closer footage of the players and the pitch so that when annotating, the representations of the important match elements would not be so easily obscured. Additionally, it would worthwhile to look into video manipulation techniques that help minimize the effects of motion blur and smearing that may occur. While higher quality footage may improve pixel representation in terms of numbers, the speed of each players' movement may still cause the usual artifacts.

C. Improve Pose Estimation

In terms of action detection, we want to eventually be able to pass sequences of poses detected in a video into another neural network that checks if it is an action of interest, i.e. a shot or a pass. However, there is one key change we would make before continuing in this path. We would also track the pose of players that are not in control of the ball. This is because during testing, we noticed that the model sometimes had issues selecting the correct player that was in control of

the ball. We suspect that this is because there is a lack of proper differentiation between players without the ball and with the ball. The only thing that contrasts the two cases is the location of the playing ball key point. However, due to the low pixel count that represents the ball, this is not a good reference point. Even on higher resolutions, the ball typically does not get bigger than 12 pixels in length and width. Combined with the fact that the frames of video are several thousand pixels in both dimensions, this is a poor way for the model to learn differences. Understanding the posing of all the players on the pitch may allow the model to better understand how the estimated poses of players performing on-ball actions compared to standard off-the-ball performances.

D. Increased Data Collection

Lastly, a natural improvement that we would make in the future is to simply collect more data. Even with the large number of annotations performed for each exploration, in comparison to other projects, this is technically a small dataset. With more examples for each of the models to train on, our accuracy for inferences can exponentially improve.

VII. CONCLUSIONS

The research has concluded that artificial intelligence and its machine learning techniques are more than capable of aiding in data analysis for sports. The statistical nuances provided by video footage can be interpreted by a neural network and efficiently relayed to a human analyst. Although some of the results may leave more to be desired, the work completed is still a successful stepping stone into the world of analysis automation.

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